Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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Recommendation

Use the latest completed assessment year, **2031-2032**, to identify teacher, course, and section patterns that deserve review before the next cycle. The stakeholder metric is **BOY/EOY score gain**: end-of-year score minus beginning-of-year score for the same public-safe student record.

The decision system does not ask whether old sections should be managed retroactively. It asks whether the most recent actual growth was materially above or below an expected-growth baseline trained only on prior years.

The baseline selected for operations is **EOY linear benchmark**. It predicts EOY score and converts that prediction into expected gain. The latest-year EOY baseline R-squared is **0.932** and the latest-year expected-gain RMSE is **4.683** points.

The latest-year gain R-squared is **0.101**. That is not treated as a failure condition because individual score gain is noisy after starting score is removed. The workflow uses the model to create a fair baseline, then makes decisions from aggregate teacher, course, and section residuals with bootstrap uncertainty checks.

Slice	Priority or watch rows	Total reviewed
Teachers	1	5
Courses	4	9
Sections	8	24

Decision	Slice	Target	N	Gap	95% CI	q
Intervention	Section	S21	12	-2.91	-5.35 to -0.55	0.200
Intervention	Section	S19	12	-2.87	-5.33 to -0.13	0.200
Intervention	Teacher	TCH-005	68	-1.24	-2.42 to -0.20	0.167
Bright spot	Section	S13	9	+2.71	0.63 to 4.71	0.040

The full decision table, including watch-list rows, is generated as reports/intervention_targets.csv.

Plain-English Method

1. Build a paired BOY/EOY growth extract from public-safe assessment records.
2. Train candidate expected-growth models on prior years only.
3. Select the operating baseline using leave-one-year-out temporal validation, with repeated CV and bootstrap checks as supporting evidence.
4. Score the latest year against that prior-year baseline.
5. Aggregate observed-minus-expected growth by teacher, course, and section.
6. Flag review targets only when the gap is large enough to matter and the uncertainty check supports follow-up.

This design separates the prediction problem from the decision problem. The prediction model estimates what growth would be expected for a similar starting profile; the review layer asks where actual growth departed from that expectation.

Direct Answers

1. The analysis covers 1,737 paired BOY/EOY records across 174 section-year groups and 5 public-safe teacher identifiers.
2. The training window is 2025-2026, 2026-2027, 2027-2028, 2028-2029, 2029-2030, 2030-2031; the action year is 2031-2032.
3. The average raw gain across the full extract is 5.72 points; the latest-year raw gain is 5.34 points.
4. The model search tested 13 candidate baselines across parametric, nonlinear, ensemble, and leakage-check families.
5. The selected baseline has temporal expected-gain RMSE 4.684 and latest-year expected-gain RMSE 4.683.
6. Teacher, course, and section flags are audit priorities. They are not automatic personnel ratings or causal claims.

Data Audit

A record enters the growth model only when the same public-safe student has valid BOY and EOY scores in the same section and teacher context. This keeps improvement tied to one instructional experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

Model Selection

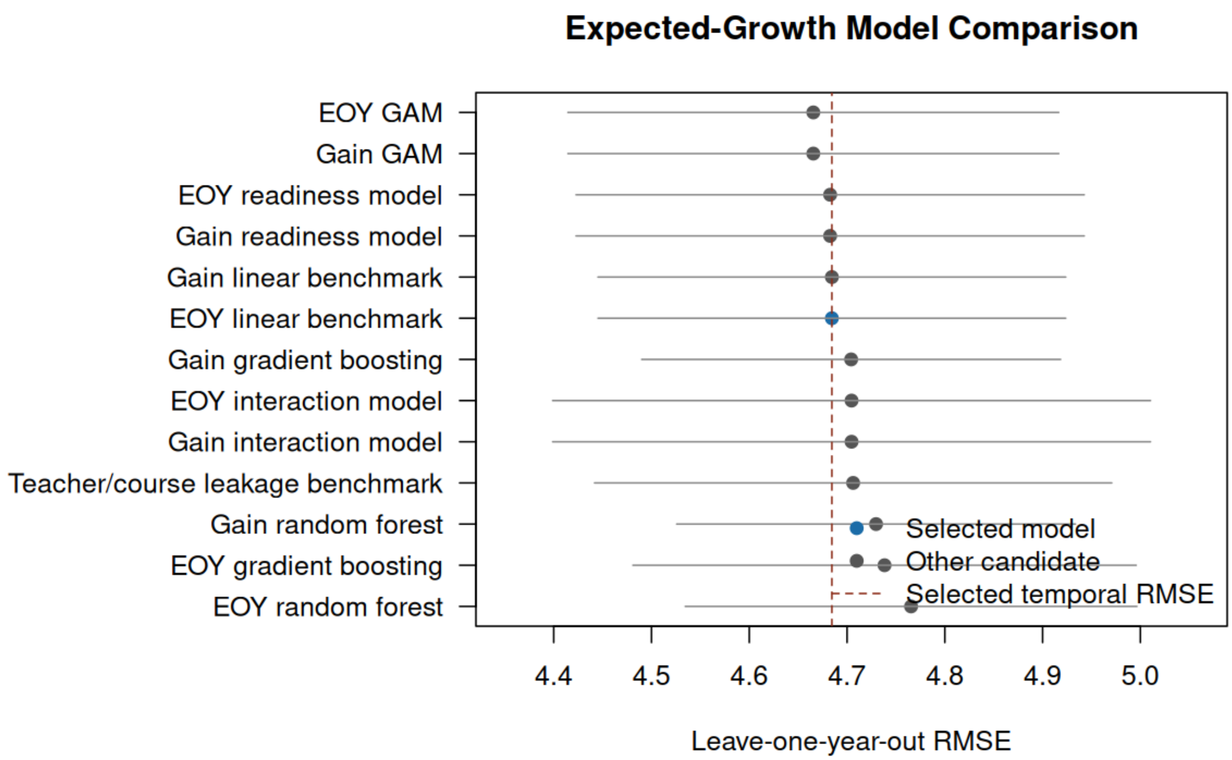
The model search used 1,485 prior-year pairs and held out 252 latest-year pairs for action-year evaluation. The primary selection metric is temporal expected-gain RMSE, not latest-year performance, so the system does not choose a model by looking at the year it later reviews.

The best raw temporal RMSE was **4.666** from **EOY GAM**. The selected model's temporal RMSE was **4.684**, a difference of 0.018 points. The simpler model was selected under the pre-specified rule because the difference was not operationally meaningful.

Family	Representative model	Decision	Temporal RMSE	Latest RMSE
Direct gain baseline	Gain readiness	Compared	4.683	4.626
Predicted EOY baseline	EOY linear	Selected	4.684	4.683
Interaction surface	EOY interaction	Compared	4.705	4.616
GAM smooths	EOY GAM	Compared	4.666	4.640
Random forest	Gain RF	Compared	4.730	4.806
Gradient boosting	Gain GBM	Compared	4.704	4.839
Teacher/course ID leakage check	Leakage check	Excluded	4.706	4.658

Model	Use	Target	Method	CV RMSE	Temporal RMSE	Latest RMSE	Latest EOY R2
EOY GAM		Predicted EOY	GAM	4.657	4.666	4.640	0.933
Gain GAM		Predicted gain	GAM	4.657	4.666	4.640	0.933
EOY readiness		Predicted EOY	Linear model	4.651	4.683	4.626	0.934
Gain readiness		Predicted gain	Linear model	4.651	4.683	4.626	0.934
EOY linear	Yes	Predicted EOY	Linear model	4.675	4.684	4.683	0.932
Gain linear		Predicted gain	Linear model	4.675	4.684	4.683	0.932
Gain GBM		Predicted gain	Gradient boosting	4.750	4.704	4.839	0.928
EOY interaction		Predicted EOY	Linear model	4.651	4.705	4.616	0.934
Leakage check		Predicted EOY	Linear model	4.680	4.706	4.658	0.933

Full model artifacts: reports/growth_model_comparison_display.csv, reports/model_temporal_validation.csv, and reports/model_bootstrap_validation.csv.


Expected-Growth Model Comparison
EOY GAM
Gain GAM
EOY readiness model
Gain readiness model
Gain linear benchmark
EOY linear benchmark
Gain gradient boosting
EOY interaction model
Gain interaction model
Teacher/course leakage benchmark
Gain random forest
EOY gradient boosting
EOY random forest
Selected model
Other candidate
Selected temporal RMSE
4.4
4.5
4.6
4.7
4.8
4.9
5.0
Leave-one-year-out RMSE

Latest-Year Review Targets

The latest-year review layer compares observed gain with expected gain for 24 section groups. The decision labels use a practical audit threshold: material gap, bootstrap interval direction, and BH-adjusted q-value for multiple-review control.

Teacher review

Target	N	Raw	Expected	Gap	CI	q	Decision
TCH-005	68	3.81	5.05	-1.24	-2.42 to -0.20	0.167	Intervention

Course review

Target	N	Raw	Expected	Gap	CI	q	Decision
AP Calc AB	49	4.31	5.47	-1.16	-2.58 to 0.05	0.180	Watch
Precalc	40	4.99	5.95	-0.95	-2.32 to 0.19	0.180	Watch
Alg 2 H	26	6.18	5.40	+0.78	-0.79 to 2.70	0.514	Watch
AP Precalc	30	6.11	4.55	+1.57	-0.07 to 2.83	0.180	Watch

Section evidence

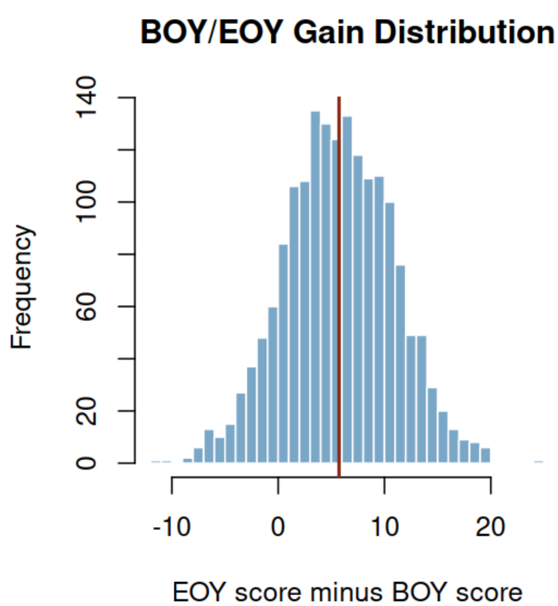
Target	N	Raw	Expected	Gap	CI	q	Decision
S21	12	3.10	6.01	-2.91	-5.35 to -0.55	0.200	Intervention
S19	12	2.41	5.28	-2.87	-5.33 to -0.13	0.200	Intervention
S13	9	7.18	4.47	+2.71	0.63 to 4.71	0.040	Bright spot
S04	9	4.01	6.45	-2.44	-6.53 to 2.42	0.582	Watch
S23	9	1.89	3.87	-1.99	-4.45 to 0.49	0.400	Watch
S16	10	4.11	5.92	-1.81	-4.73 to 0.62	0.582	Watch
S05	11	4.00	5.28	-1.28	-3.86 to 1.69	0.686	Watch
S14	10	4.81	5.76	-0.95	-3.93 to 2.11	0.747	Watch

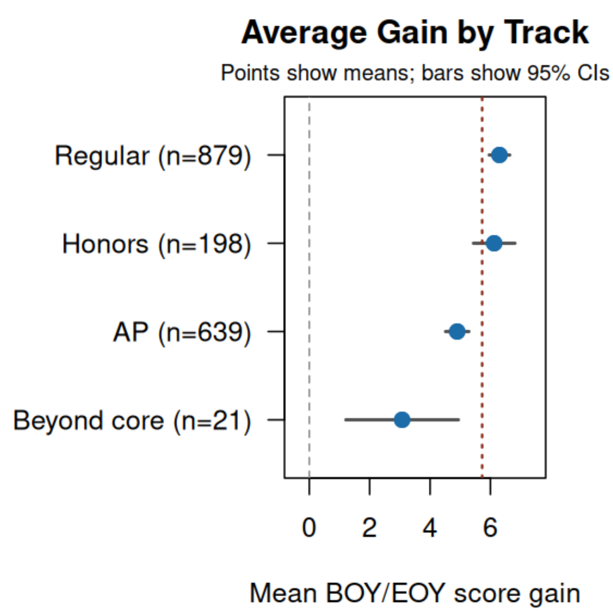
Full review tables: reports/latest_teacher_review.csv, reports/latest_course_review.csv, and reports/latest_section_review.csv.

Section Evidence

Raw section improvement is useful for communication, but it is not the final comparison. A section can show positive raw gain and still fall below expected growth if its starting profile suggested a larger increase.

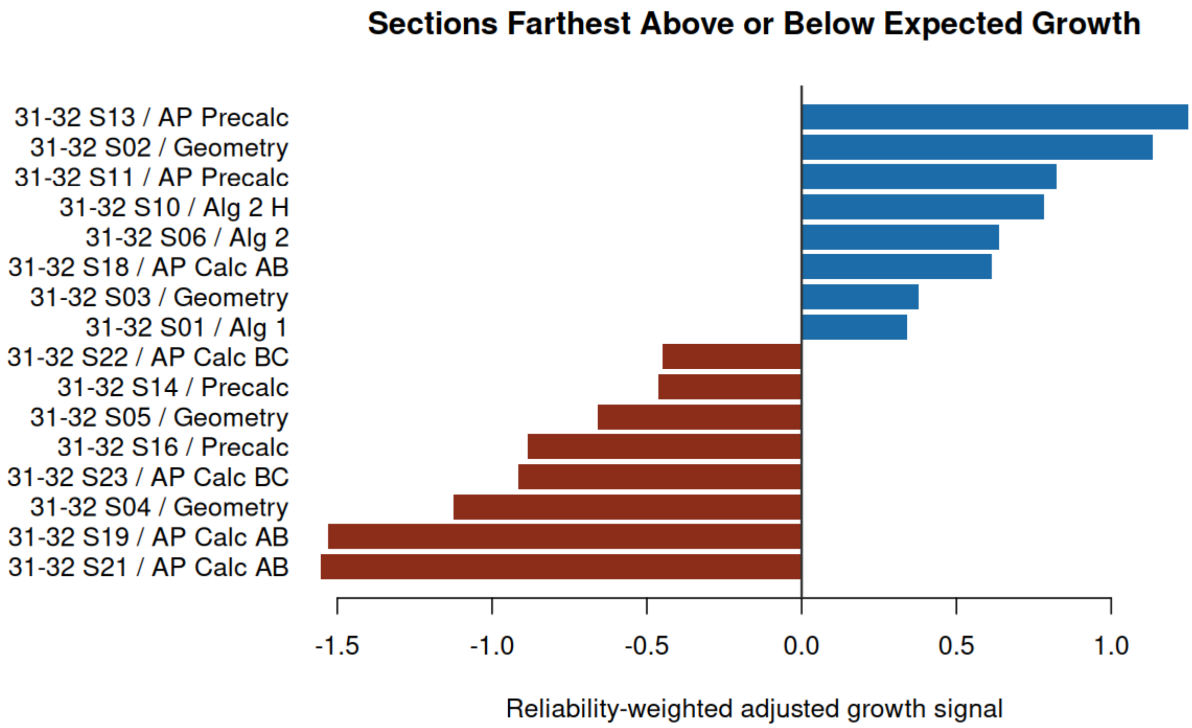
Section	Course	N	BOY	EOY	Gain	95% CI	p-value
S13	AP Precalc	9	63.5	70.7	7.18	5.20 to 9.16	<0.001
S02	Geometry	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
S11	AP Precalc	10	69.0	74.7	5.67	1.52 to 9.83	0.013
S10	Alg 2 H	12	63.1	70.0	6.90	3.98 to 9.83	<0.001
S06	Alg 2	11	41.0	48.6	7.63	4.97 to 10.30	<0.001
S18	AP Calc AB	13	49.4	56.1	6.69	3.06 to 10.31	0.002
S03	Geometry	9	43.2	50.2	7.02	3.12 to 10.92	0.003
S01	Alg 1	13	42.1	49.2	7.13	4.19 to 10.08	<0.001





The adjusted section signal is observed gain minus expected gain, reliability-weighted toward zero for smaller sections.

Section	Teacher	Course	N	Raw	Expected	Signal	Result
S13	TCH-004	AP Precalc	9	7.18	4.47	+1.25	Above expected
S02	TCH-002	Geometry	10	9.02	6.70	+1.13	Within expected range
S11	TCH-004	AP Precalc	10	5.67	3.98	+0.83	Within expected range
S10	TCH-003	Alg 2 H	12	6.90	5.44	+0.78	Within expected range
S16	TCH-004	Precalc	10	4.11	5.92	-0.88	Within expected range
S05	TCH-001	Geometry	11	4.00	5.28	-0.66	Within expected range
S14	TCH-003	Precalc	10	4.81	5.76	-0.46	Within expected range
S22	TCH-005	AP Calc BC	10	3.10	4.02	-0.45	Within expected range



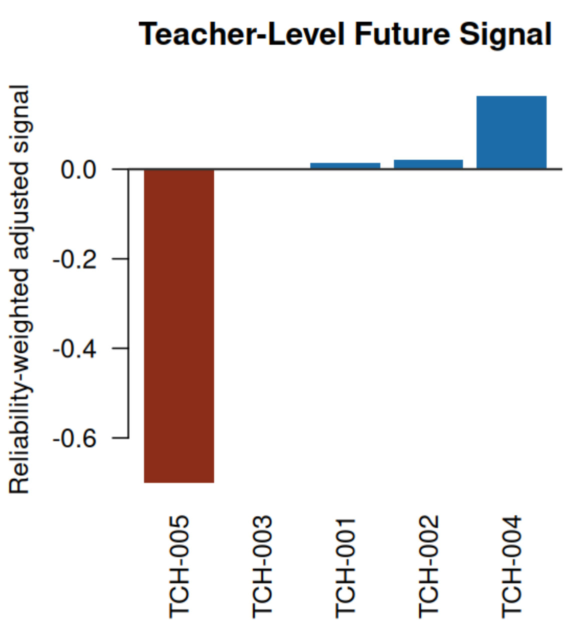
The full section signal table is generated as reports/section_adjusted_signals.csv.

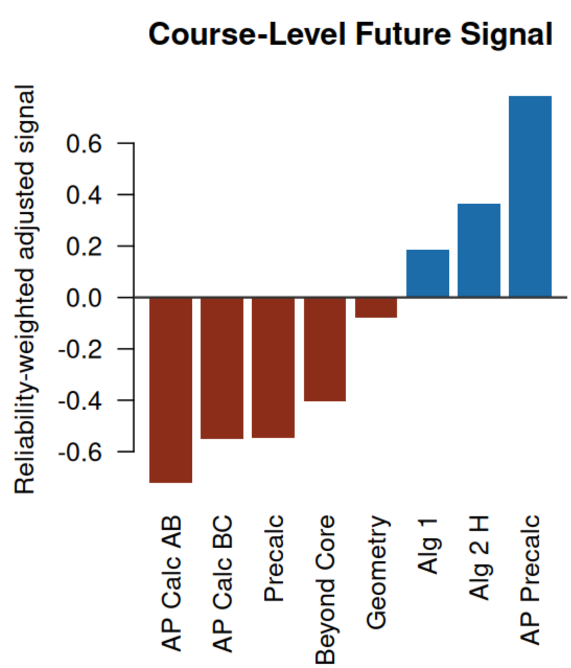
Teacher and Course Summaries

These summaries aggregate the latest-year evidence into planning views. They support review conversations about pacing, curriculum alignment, attendance mix, and transferable practices. They should not be read as standalone personnel scores.

Teacher	Sections	Records	Raw	Expected	Signal
TCH-004	5	43	5.21	4.85	+0.16
TCH-002	5	52	6.60	6.56	+0.02
TCH-001	3	33	6.06	6.02	+0.01
TCH-003	5	56	5.70	5.70	0.00
TCH-005	6	68	3.81	5.05	-0.70

Course	Sections	Records	Raw	Expected	Signal
AP Precalc	3	30	6.11	4.55	+0.78
Alg 2 H	2	26	6.18	5.40	+0.36
Alg 1	1	13	7.13	6.52	+0.19
Alg 2	3	33	6.57	6.54	+0.01
Geometry	4	39	5.99	6.13	-0.08
Beyond Core	1	3	-0.11	4.35	-0.41
Precalc	4	40	4.99	5.95	-0.55
AP Calc BC	2	19	2.52	3.95	-0.55
AP Calc AB	4	49	4.31	5.47	-0.72





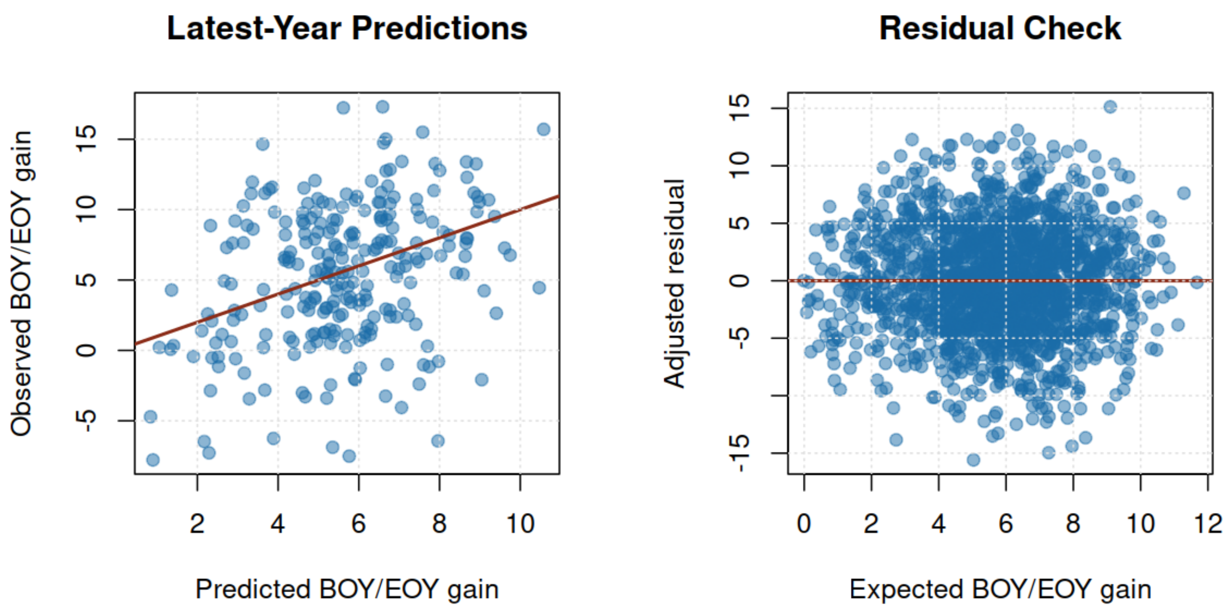
Diagnostics and Sensitivity

The baseline is strong for final-score expectation and weaker for individual gain variation. That pattern is expected: BOY score explains much of EOY score, while individual improvement contains more unobserved classroom, attendance, and assessment noise.

Diagnostic	Estimate	Interpretation
Latest-year expected-gain RMSE	4.683	Typical out-of-sample prediction error on latest-year BOY/EOY gain
Latest-year expected-gain R-squared	0.101	Share of latest-year gain variation explained by the expected-growth model
Latest-year EOY R-squared	0.932	Share of latest-year EOY score variation explained by the baseline
Latest-year residual mean	-0.258	Near 0 means expected gain is centered in the action year
Latest-year residual SD	4.686	Latest-year residual spread used for slice uncertainty

Metric	Estimate	95% interval
Expected-gain RMSE	4.683	4.270 to 5.104
Expected-gain MAE	3.728	3.375 to 4.072
Expected-gain R-squared	0.101	-0.007 to 0.186
EOY RMSE	4.683	4.270 to 5.104
EOY R-squared	0.932	0.916 to 0.944

Measure	Value
Training paired records	1,485
Latest-year paired records	252
Latest-year section-year groups	24
Latest-year mean raw BOY/EOY gain	5.34
Latest-year raw-vs-adjusted rank correlation	0.875
Latest-year top-10 overlap, raw vs adjusted ranking	70.0%



Technical Appendix

The operating model excludes teacher IDs, course IDs, and section IDs. A leakage benchmark is reported only to show what would happen if persistent IDs were included in the baseline; it is not used for review because it would absorb the teacher/course patterns the decision layer is designed to detect.

Package	Installed
mgcv	Available
randomForest	Available
gbm	Available

Metric	Value
Selected model	EOY linear benchmark
Selected target strategy	Predicted EOY
Selected method	Linear model
Selection rule	Lowest temporal-validation expected-gain RMSE among operational candidates; ties within 1% choose simpler model
Training paired records	1,485
Latest-year action paired records	252
Training years	2025-2026, 2026-2027, 2027-2028, 2028-2029, 2029-2030, 2030-2031
Action year	2031-2032
Candidate models tested	13
Operational candidates tested	12
Excluded leakage benchmarks	1
Repeated CV folds	5

Metric	Value
Repeated CV repeats	3
Temporal expected-gain RMSE	4.684
Temporal expected-gain R-squared	0.153
Temporal EOY R-squared	0.937
Latest-year expected-gain RMSE	4.683
Latest-year expected-gain R-squared	0.101
Latest-year EOY R-squared	0.932

The full selected-model metrics table is generated as reports/growth_final_metrics.csv.

Conclusion

The project should be read as a statistical decision-support system, not as a simple prediction demo. The strongest business value is the workflow: choose a validated expected-growth baseline, compare latest actual growth to that baseline, quantify uncertainty by slice, and translate the evidence into review priorities.

The recommended stakeholder action is to review the flagged teacher, course, and section patterns before the next assessment cycle. Priority targets deserve support or investigation; positive anomalies deserve study for transferable practices; watch-list rows deserve context review before escalation.

The important limitation is that the data are public-safe and generalized from an assessment workflow. The outputs demonstrate the analysis pattern and should not be used as real student, teacher, or personnel decisions.

Reproducibility

Rebuild the full evidence packet with `make all`. The pipeline uses the included public-safe extract and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.